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Course: Calculus Beta

Assignment: MyLab Math - Uge 12

Find f_x , f_y , f_z .

$$f(x,y,z) = 5x - \sqrt{7y^2 + 2z^2}$$

The symbol f_x means the partial derivative of f with respect to x . To find this derivative, treat y and z as constants and differentiate the given function f with respect to x in the usual way.

To begin, differentiate the first term in the function with respect to x .

$$\begin{aligned} f_x &= \frac{\partial}{\partial x} 5x - \frac{\partial}{\partial x} \sqrt{7y^2 + 2z^2} \\ &= 5 - \frac{\partial}{\partial x} \sqrt{7y^2 + 2z^2} \end{aligned}$$

Then differentiate the second term with respect to x .

$$\begin{aligned} f_x &= 5 - \frac{\partial}{\partial x} \sqrt{7y^2 + 2z^2} \\ &= 5 - 0 \\ &= 5 \end{aligned}$$

The symbol f_y means the partial derivative of f with respect to y . To find this derivative, treat x and z as constants and differentiate the given function f with respect to y in the usual way.

To begin, differentiate the first term in the function with respect to y .

$$\begin{aligned} f_y &= \frac{\partial}{\partial y} 5x - \frac{\partial}{\partial y} \sqrt{7y^2 + 2z^2} \\ &= 0 - \frac{\partial}{\partial y} \sqrt{7y^2 + 2z^2} \end{aligned}$$

Now differentiate the second term by treating it as the square root of a function of y and using the Chain Rule.

$$\begin{aligned} f_y &= 0 - \frac{\partial}{\partial y} \sqrt{7y^2 + 2z^2} \\ &= - \frac{\partial}{\partial y} \sqrt{7y^2 + 2z^2} \\ &= - \frac{1}{2\sqrt{7y^2 + 2z^2}} \frac{\partial}{\partial y} (7y^2 + 2z^2) \end{aligned}$$

Then evaluate the partial derivative in the result.

$$\begin{aligned} f_y &= - \frac{1}{2\sqrt{7y^2 + 2z^2}} \frac{\partial}{\partial y} (7y^2 + 2z^2) \\ &= - \frac{1}{2\sqrt{7y^2 + 2z^2}} (14y) && \text{Evaluate.} \\ &= - \frac{7y}{\sqrt{7y^2 + 2z^2}} && \text{Simplify.} \end{aligned}$$

The symbol f_z means the partial derivative of f with respect to z . To find this derivative, treat x and y as constants and differentiate the given function f with respect to z in the usual way.

To begin, differentiate the first term in the function with respect to z and using the Chain Rule.

$$\begin{aligned} f_z &= \frac{\partial}{\partial z} 5x - \frac{\partial}{\partial z} \sqrt{7y^2 + 2z^2} \\ &= 0 - \frac{\partial}{\partial z} \sqrt{7y^2 + 2z^2} \end{aligned}$$

Now differentiate the second term by treating it as the square root of a function of z .

$$\begin{aligned}
 f_z &= 0 - \frac{\partial}{\partial z} \sqrt{7y^2 + 2z^2} \\
 &= - \frac{\partial}{\partial z} \sqrt{7y^2 + 2z^2} \\
 &= - \frac{1}{2\sqrt{7y^2 + 2z^2}} \frac{\partial}{\partial z} (7y^2 + 2z^2)
 \end{aligned}$$

Then evaluate the partial derivative in the result.

$$\begin{aligned}
 f_z &= - \frac{1}{2\sqrt{7y^2 + 2z^2}} \frac{\partial}{\partial z} (7y^2 + 2z^2) \\
 &= - \frac{1}{2\sqrt{7y^2 + 2z^2}} (4z) && \text{Evaluate.} \\
 &= - \frac{2z}{\sqrt{7y^2 + 2z^2}} && \text{Simplify.}
 \end{aligned}$$

Thus, the partial derivatives are shown below.

$$f_x = 5 \qquad f_y = - \frac{7y}{\sqrt{7y^2 + 2z^2}} \qquad f_z = - \frac{2z}{\sqrt{7y^2 + 2z^2}}$$